

1)

```
mysql> USE student;
```

Database changed

```
mysql> SELECT s.name, d.dept_name
```

```
-> FROM student s, department d;
```

```
+-----+-----+
| name | dept_name |
+-----+-----+
| Arun | Biotechnology |
| Arun | Computer Science |
| Arun | Bioinformatics |
| Ravi | Biotechnology |
| Ravi | Computer Science |
| Ravi | Bioinformatics |
| Priya | Biotechnology |
| Priya | Computer Science |
| Priya | Bioinformatics |
| Kavi | Biotechnology |
| Kavi | Computer Science |
| Kavi | Bioinformatics |
| Meena | Biotechnology |
| Meena | Computer Science |
| Meena | Bioinformatics |
```

```
+-----+-----+
```

15 rows in set (0.00 sec)

```
mysql> SELECT s.name, d.dept_name
```

```
-> FROM student s
```

```
-> JOIN department d
```

```
-> ON s.dept_id = d.dept_id;
```

```
+-----+-----+
```

name	dept_name
Arun	Bioinformatics
Priya	Bioinformatics
Meena	Bioinformatics
Ravi	Computer Science
Kavi	Biotechnology

2)

```
mysql> SELECT s.name, d.dept_name
```

```
-> FROM student s
```

```
-> JOIN department d
```

```
-> ON s.dept_id = d.dept_id;
```

name	dept_name
Arun	Bioinformatics
Priya	Bioinformatics
Meena	Bioinformatics
Ravi	Computer Science
Kavi	Biotechnology

5 rows in set (0.00 sec)

```
mysql> SELECT s.name
```

```
-> FROM student s
```

```
-> WHERE EXISTS (
```

```
->   SELECT 1
```

```
->   FROM department d
```

```
-> );
```

name

name

--

Arun

Ravi

Priya

Kavi

Meena

--

3)

```
mysql> CREATE VIEW Student_Department AS
```

```
-> SELECT s.name, d.dept_name
```

```
-> FROM student s
```

```
-> JOIN department d;
```

ERROR 1050 (42S01): Table 'Student_Department' already exists

```
mysql> CREATE OR REPLACE VIEW Student_Department AS
```

```
-> SELECT s.student_id, s.name, d.dept_name
```

```
-> FROM student s
```

```
-> JOIN department d
```

```
-> ON s.dept_id = d.dept_id;
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> SELECT * FROM Student_Department;
```

--	--	--

student_id	name	dept_name
------------	------	-----------

--	--	--

101	Arun	Bioinformatics
-----	------	----------------

103	Priya	Bioinformatics
-----	-------	----------------

105	Meena	Bioinformatics
-----	-------	----------------

102	Ravi	Computer Science
-----	------	------------------

104	Kavi	Biotechnology
-----	------	---------------

--	--	--

4)

```
mysql> SELECT *
```

```
-> FROM student
```

```
-> NATURAL JOIN department;
```

```
+-----+-----+-----+-----+
| dept_id | student_id | name | dept_name |
+-----+-----+-----+-----+
| 10 | 101 | Arun | Bioinformatics |
| 10 | 103 | Priya | Bioinformatics |
| 10 | 105 | Meena | Bioinformatics |
| 20 | 102 | Ravi | Computer Science |
| 30 | 104 | Kavi | Biotechnology |
+-----+-----+-----+-----+
```

5)

```
mysql> INSERT INTO student (student_id, name, dept_id)
```

```
-> VALUES (105, 'Meena', 50);
```

```
ERROR 1062 (23000): Duplicate entry '105' for key 'student.PRIMARY'
```

```
mysql> INSERT INTO student (student_id, name, dept_id)
```

```
-> VALUES (105, 'Meena', 10);
```

```
ERROR 1062 (23000): Duplicate entry '105' for key 'student.PRIMARY'
```

```
mysql> SELECT * FROM student;
```

```
+-----+-----+-----+
| student_id | name | dept_id |
+-----+-----+-----+
| 101 | Arun | 10 |
| 102 | Ravi | 20 |
| 103 | Priya | 10 |
| 104 | Kavi | 30 |
| 105 | Meena | 10 |
+-----+-----+-----+
```

6)

```
mysql> SELECT s.name,
-> (SELECT d.dept_name
-> FROM department d
-> WHERE d.dept_id = s.dept_id) AS dept_name
-> FROM student s;
```

```
+-----+-----+
| name | dept_name |
+-----+-----+
| Arun | Bioinformatics |
| Ravi | Computer Science |
| Priya | Bioinformatics |
| Kavi | Biotechnology |
| Meena | Bioinformatics |
+-----+-----+
```

5 rows in set (0.00 sec)

```
mysql> SELECT s.name, d.dept_name
-> FROM student s
-> JOIN department d
-> ON s.dept_id = d.dept_id;
```

```
+-----+-----+
| name | dept_name |
+-----+-----+
| Arun | Bioinformatics |
| Priya | Bioinformatics |
| Meena | Bioinformatics |
| Ravi | Computer Science |
| Kavi | Biotechnology |
+-----+-----+
```

7)

```
mysql> SELECT dept_id, name, COUNT(*)
```

-> FROM student

-> GROUP BY dept_id;

ERROR 1055 (42000): Expression #2 of SELECT list is not in GROUP BY clause and contains nonaggregated column 'student.student.name' which is not functionally dependent on columns in GROUP BY clause; this is incompatible with sql_mode=only_full_group_by

mysql> SELECT dept_id, COUNT(*) AS student_count

-> FROM student

-> GROUP BY dept_id;

```
+-----+-----+
| dept_id | student_count |
+-----+-----+
| 10 | 3 |
| 20 | 1 |
| 30 | 1 |
+-----+-----+
```

8)

mysql> UPDATE student

-> SET dept_id = 20;

Query OK, 4 rows affected (0.00 sec)

Rows matched: 5 Changed: 4 Warnings: 0

mysql> UPDATE student

-> SET dept_id = 20

-> WHERE student_id = 101;

Query OK, 0 rows affected (0.00 sec)

Rows matched: 1 Changed: 0 Warnings: 0

mysql> SELECT * FROM student;

```
+-----+-----+-----+
| student_id | name | dept_id |
+-----+-----+-----+
| 101 | Arun | 20 |
```

	102	Ravi		20	
	103	Priya		20	
	104	Kavi		20	
	105	Meena		20	

+-----+-----+-----+

9)

mysql> SELECT s.name

-> FROM student s

-> WHERE s.dept_id IN (

-> SELECT d.dept_id

-> FROM department d

-> WHERE d.dept_name = 'Bioinformatics'

->);

Empty set (0.00 sec)

mysql> SELECT s.name, d.dept_name

-> FROM student s

-> JOIN department d

-> ON s.dept_id = d.dept_id

-> WHERE d.dept_name = 'Bioinformatics';

Empty set (0.00 sec)

mysql> SELECT * FROM student;

+-----+-----+-----+

student_id	name	dept_id	
------------	------	---------	--

+-----+-----+-----+

	101	Arun		20	
--	-----	------	--	----	--

	102	Ravi		20	
--	-----	------	--	----	--

	103	Priya		20	
--	-----	-------	--	----	--

	104	Kavi		20	
--	-----	------	--	----	--

	105	Meena		20	
--	-----	-------	--	----	--

+-----+-----+-----+

10)

```
mysql> { s | Student(s)  $\wedge \forall d$  (Department(d)  $\rightarrow$  s.dept_id  $\neq$  d.dept_id) }
```

```
-> ;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '{ s | Student(s) ? ?d (Department(d) ? s.dept_id ? d.dept_id) }' at line 1

```
mysql> { s | Student(s)  $\wedge \exists d$  (Department(d)  $\wedge$  s.dept_id = d.dept_id) }
```

```
-> ;
```

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '{ s | Student(s) ? ?d (Department(d) ? s.dept_id = d.dept_id) }' at line 1

```
mysql> SELECT s.*
```

```
-> FROM student s
```

```
-> WHERE EXISTS (
```

```
->   SELECT 1
```

```
->   FROM department d
```

```
->   WHERE d.dept_id = s.dept_id
```

```
-> );
```

+-----+-----+-----+

| student_id | name | dept_id |

+-----+-----+-----+

| 101 | Arun | 20 |

| 102 | Ravi | 20 |

| 103 | Priya | 20 |

| 104 | Kavi | 20 |

| 105 | Meena | 20 |

+-----+-----+-----+